

Question 1

Question text

1. The first step of machine learning is_____.

Select one:

- a. Modeling
- b. Testing
- c. Data acquisition
- d. Data representation
- e. None

Feedback

The correct answer is: Data acquisition

Question 2

10. From the list of transition functions given below, select the transition function for F-NFA.

Select one:

- a. $A(q_1, x), q_2$, where $q_1, q_2 \in Q$ and $x \in \Sigma$.
- b. $B(q_1, x), q_2^+$, where $q_1, q_2 \in Q$ and $x \in \Sigma$.
- c. $C(q_1, x), q_2^*$, where $q_1, q_2 \in Q$ and $x \in \Sigma$.
- d. $D(q_1, x \cup \epsilon) \rightarrow q_2^*$, where $q_1, q_2 \in Q$ and $x \in \Sigma$.
- e. All except A.

Feedback

The correct answer is: $D(q_1, x \cup \epsilon) \rightarrow q_2^*$, where $q_1, q_2 \in Q$ and $x \in \Sigma$.

Question 3

100. The opposing forces to Integrity, Confidentiality, and Availability are?

Select one:

- a. Disclosure, Alteration, and destruction
- b. Disclosure, destruction, and alteration
- c. Alteration, destruction, and disclosure
- d. Alteration, disclosure, and destruction

Feedback

The correct answer is: Alteration, destruction, and disclosure

Question 4

101. Which one of the following is not a kind of passive network security attack?

Select one:

- a. Traffic analysis
- b. Release of message content
- c. Wire tapping
- d. Session hijacking

Feedback

The correct answer is: Wire tapping

Question 5

102. Which one of the following is an example of a block cipher and Asymmetric cryptosystem?

Select one:

- a. DES
- b. RC4
- c. RSA
- d. None of the above

Feedback

The correct answer is: None of the above

Question 6

103. Which one of the following cryptographic algorithms is odd?

Select one:

- a. RSA
- b. DES
- c. RC4
- d. 3DES

Question 7

104. Which of the following statement is true about network security?

Select one:

- a. It is a set of rules and configurations designed to protect the integrity, confidentiality and accessibility of computer networks and data using both software and hardware technologies.
- b. Every organization, regardless of size, industry or infrastructure, requires a degree of network security solutions in place to protect it from the ever-growing landscape of cyber threats in the world today
- c. Today's network architecture is complex and we faced with a dangerous environment that is always changing and attackers that are always trying to find and exploit vulnerabilities are highly increasing.
- d. All of the above

Feedback

The correct answer is: Today's network architecture is complex and we faced with a dangerous environment that is always changing and attackers that are always trying to find and exploit vulnerabilities are highly increasing.

Question 8

105. Imagine, as a user of Hawassa University's SIS system, you are unable to login using your correct username and password. Which computer security goal is not fulfilled?

Select one:

- a. Integrity
- b. Availability
- c. Authenticity
- d. Confidentiality

Feedback

The correct answer is: Availability

Question 9

106. Which one of the following services is not provided by SET?

Select one:

- a. Confidentiality of information
- b. Accountability
- c. Authentication
- d. Integrity of data

Feedback

The correct answer is: Accountability

Question 10

107. Which of the following statements are not correct?

Select one:

- a. In symmetric cryptography, only single key is shared between sender and receiver.
- b. DES is a symmetric cryptography algorithm which groups the message into 64 bits to encrypt.
- c. Rail Fence cipher is a public-key cryptographic algorithm.
- d. PGP provides confidentiality and authentication service for electronic mail application.

Feedback

The correct answer is: Rail Fence cipher is a public-key cryptographic algorithm.

Question 11

108. What is an umbrella term for managing access to a network?

Select one:

- a. AR
- b. NAS
- c. NAC
- d. All of the above

Feedback

The correct answer is: NAS

Question 12

109. Which one of the following is used to stop traffic or permit only specified traffic while stopping all other traffic on their network?

Select one:

- a. Default gateway
- b. ACL
- c. Routing table

- d. All of the above
- e. None of the above

Feedback

The correct answer is: ACL

Question 13

11. What is dead state?

Select one:

- a. A state that can lead the transition to final state.
- b. A state that cannot lead the transition to final state.
- c. A state which is unreachable from initial state.
- d. A state which can be reached from initial state.
- e. All Except A.

Feedback

The correct answer is: A state that cannot lead the transition to final state.

Question 14

110. Which one of the following router component is used to contain the operating system?

Select one:

- a. NVRAM
- b. ROM
- c. RAM
- d. A and C
- e. None of the above

Feedback

The correct answer is: None of the above

Question 15

111. What is a routing protocol that uses number of hops as its metric?

Select one:

- a. EIGRP

- b. OSPF
- c. RIP
- d. All of the above
- e. None of the above

Feedback

The correct answer is: RIP

Question 16

112. Which one of the following routing protocol uses unequal cost load balancing?

Select one:

- a. EIGRP
- b. RIP
- c. IS-IS
- d. None of the above

Question 17

113. What is an address that is not changed when a packet is forwarded from router to router?

Select one:

- a. IP address
- b. Port number
- c. MAC address
- d. None of the above
- e. All of the above

Feedback

The correct answer is: IP address

Question 18

114. What is the interface of a router that is used to connect to the LAN?

Select one:

- a. Serial interface
- b. Fast Ethernet interface
- c. Console interface

- d. A and B
- e. B and C

Feedback

The correct answer is: Fast Ethernet interface

Question 19

115. What is the command that is used to activate an interface?

Select one:

- a. ipconfig
- b. ip route
- c. shutdown
- d. All of the above
- e. None of the above

Feedback

The correct answer is: None of the above

Question 20

116. What is the type of ACL in which packets are processed before they are routed to the outbound interface?

Select one:

- a. Outbound ACL
- b. ACL loop
- c. Inbound ACL
- d. None of the above
- e. All of the above

Feedback

The correct answer is: Inbound ACL

Question 21

117. Which one of the following is not included in the command of static routing?

Select one:

- a. Subnet mask

- b. Wildcard mask
- c. Destination ip address
- d. None of the above
- e. All of the above

Feedback

The correct answer is: Wildcard mask

Question 22

118. What is the command used to save running configuration file?

Select one:

- a. Router#copy startup-config running-config
- b. Router(config)#copy startup-config running-config
- c. Router(config-line)#copy startup-config running-config
- d. All of the above
- e. None of the above

Feedback

The correct answer is: None of the above

Question 23

119. Which one of the following Structured Query Languages (SQL) is used to create a database?

Select one:

- a. DML
- b. DDL
- c. DCL
- d. TCL

Feedback

The correct answer is: DDL

Question 24

12. Which of the following equality if false?

Select one:

- a. $(L^*)^* = L^*$
- b. $B^* = \emptyset$
- c. $C^* = *$
- d. $L^+ = LL^*$
- e. $L^* = L + L^+$

Feedback

The correct answer is: $B^* = \emptyset$

Question 25

120. Which of the following is a constraint between two sets of attributes from the database?

Select one:

- a. Normalization
- b. functional dependency
- c. Denormalization
- d. All

Feedback

The correct answer is: functional dependency

Question 26

13. According to the Church-Turing thesis, a problem is solvable if and only if _____?

Select one:

- a. It converted into equivalent Turing machine problem.
- b. A Turing machine halts for all inputs.
- c. A Turing machine halts only when it accepts the input.
- d. A and B
- e. B and C

Feedback

The correct answer is: A and B

Question 27

14. Which of the following language cannot be recognized using a finite state automaton?

Select one:

- a. $0^n 1^n$ for $n \geq 1$
- b. $0^n 1^n 0^n$ for $n \geq 1$
- c. WWR for $W=\{0,1\}$ and $R \geq 1$
- d. $(0^* 1^*)^*$
- e. All of the above.

Feedback

The correct answer is: $0^n 1^n 0^n$ for $n \geq 1$

Question 28

15. A context free grammar is ambiguous if and only if _____?

Select one:

- a. It has one rightmost and one leftmost derivation tree for a string.
- b. It has more than one rightmost derivation tree for a string.
- c. It has more than one leftmost derivation tree for a string.
- d. B and C
- e. None of the above.

Feedback

The correct answer is: B and C

Question 29

16. Which automata-memory combination is wrong?

Select one:

- a. Finite automata – queue
- b. Turing machine – tape
- c. Linear bound automata – tape
- d. Pushdown automata – stack
- e. A and C

Question 30

17. Which statement is true according to the Noam Chomsky hierarchy of grammars?

Select one:

- a. Regular grammars are subsets of context-free grammars.
- b. Context-free grammars are subsets of context-sensitive grammars.
- c. Context-free grammars are subsets of unrestricted grammars.
- d. Regular grammars are subsets of context-sensitive grammars.
- e. All of the above.

Feedback

The correct answer is: All of the above.

Question 31

18. Choose the right arrangement in increasing order computing capability of the following theoretical computing machines.

Select one:

- a. Pushdown automata, finite automata, linear bound automata then Turing machine.
- b. finite automata, linear bound automata, pushdown automata then Turing machine.
- c. Turing machine, linear bound automata, pushdown automata then finite automata.
- d. Linear bound automata, pushdown automata, finite automata then Turing machine.
- e. Finite automata, pushdown automata, linear bound automata then Turing machine.

Feedback

The correct answer is: Finite automata, pushdown automata, linear bound automata then Turing machine.

Question 32

19. Among the following addressing modes, which addressing mode is more suitable to array assignment?

Select one:

- a. Immediate AM
- b. Direct AM

- c. Index AM
- d. Base-register AM

Feedback

The correct answer is: Index AM

Question 33

2. which one of the following is challenge of machine learning?

Select one:

- a. Lack of data
- b. Quality of the dataset
- c. Limitations of processing machines and chips
- d. All
- e. None

Feedback

The correct answer is: All

Question 34

20. Which statement is correct for recognizing identifier lexemes from the input buffer?

Select one:

- a. Look forward until incidence of white-space.
- b. Look forward until incidence of non-alphanumeric characters.
- c. Look forward until incidence of numeric characters.
- d. Look forward until incidence of alphabet characters.
- e. All except A.

Feedback

The correct answer is: Look forward until incidence of non-alphanumeric characters.

Question 35

21. Assume we are designing a compiler for AAA language, and we need to design it to work on three different architectures namely Windows, Linux and Macintosh

operating systems. Which phases of the compiler can be reused in all three architectures?

Select one:

- a. Lexical, Syntax and Semantic analysis
- b. Intermediate code generation, Code optimization and Code generation
- c. Semantic analysis, Code optimization and Code generation
- d. Intermediate code generation and code optimization
- e. Only lexical analysis

Feedback

The correct answer is: Lexical, Syntax and Semantic analysis

Question 36

22. Which of the following protocol is used for E-Mail Traffic?

Select one:

- a. ARP
- b. DMS
- c. SMTP
- d. IM

Feedback

The correct answer is: SMTP

Question 37

23. The transmission signal coding method that use three levels is called_____

Select one:

- a. NRZ
- b. Bipolar AMI
- c. Manchester
- d. Binary

Feedback

The correct answer is: Bipolar AMI

Question 38

24. Which type of topology is best suited for large businesses which must carefully control and coordinate the operation of distributed branch outlets?

Select one:

- a. Ring
- b. Local area
- c. Hierarchical
- d. Star

Feedback

The correct answer is: Star

Question 39

25. What kind of transmission medium is most appropriate to carry data in a computer network that is exposed to electrical interferences?

Select one:

- a. Unshielded twisted pair
- b. Optical fiber
- c. Coaxial cable
- d. Microwave

Feedback

The correct answer is: Coaxial cable

Question 40

26. The IP network 192.168.50.0 is to be divided into 8 equal sized subnets. Which of the following subnet masks can be used for the above requirement?

Select one:

- a. 255.255.243.240.
- b. 255.255.0.0
- c. 255.255.255.0
- d. 255.255.2.55.224

Feedback

The correct answer is: 255.255.2.55.224

Question 41

27. What is the size of the destination port in the UDP protocol?

Select one:

- a. 8 bits
- b. 16 bits
- c. 20 bits
- d. 32 bits

Feedback

The correct answer is: 16 bits

Question 42

28. Which of the following is the purpose of computer network?

Select one:

- a. Sharing of software
- b. Sharing of hardware resource
- c. Sharing of data
- d. All of the above

Feedback

The correct answer is: All of the above

Question 43

29. A routing information of a router that is set by the network administrator is called _____

Select one:

- a. Fixed routing
- b. Static routing
- c. Dynamic routing
- d. Automatic routing

Feedback

The correct answer is: Static routing

Question 44

3. What is state space in the field of Artificial Intelligence?

Select one:

- a. The whole problem
- b. Your definition to a problem
- c. Problem you design
- d. Representation of a problem using states and actions
- e. None

Feedback

The correct answer is: Representation of a problem using states and actions

Question 45

30. What can be used to store one or more bits of data, which can accept and transfer information serially?

Select one:

- a. Parallel registers
- b. Counters
- c. Shift registers
- d. None

Feedback

The correct answer is: Shift registers

Question 46

31. Counter that holds addresses of next fetched instruction is called:

Select one:

- a. sequence control register
- b. temporary register
- c. program counter
- d. both A and B

Feedback

The correct answer is: program counter

Question 47

32. The primary purpose of a three-state buffer is usually:

Select one:

- a. to provide isolation between the input device and the data bus
- b. to provide the sink or source current required by any device connected to its output without loading down the output device
- c. temporary data storage
- d. to control data flow

Feedback

The correct answer is: to provide isolation between the input device and the data bus

Question 48

33. Which language is termed as the symbolic depiction used for indicating the series:

Select one:

- a. Random transfer language
- b. Arithmetic transfer language
- c. Register transfer language
- d. All of these

Feedback

The correct answer is: Register transfer language

Question 49

34. The register is a type of _____.

Select one:

- a. Combinational circuit
- b. Sequential circuit
- c. CPU
- d. Latches

Feedback

The correct answer is: Sequential circuit

Question 50

35. In the direct address mode:

Select one:

- a. The effective address is equal to the address part of the instruction
- b. The content of the program counter is added to address part of the instruction
- c. A memory is addressed by a register
- d. Address in the instruction points to location of the effective address

Feedback

The correct answer is: The effective address is equal to the address part of the instruction

Question 51

36. An instruction used to set the carry flag in a computer can be classified as

Select one:

- a. Data transfer
- b. Process control
- c. Logical
- d. None of these

Feedback

The correct answer is: Logical

Question 52

37. Assume that a digital computer has a common bus system for 8 registers of 16 bits each. Then, how many multiplexers and selection inputs in each multiplexer are there, respectively?

Select one:

- a. 8 and 3
- b. 16 and 3
- c. 3 and 8
- d. 3 and 16
- e. None

Feedback

The correct answer is: 16 and 3

Question 53

38. Dijkstra's banking algorithm in an operating system solves the problem of:

Select one:

- a. deadlock avoidance
- b. deadlock recovery
- c. mutual exclusion
- d. context switching

Feedback

The correct answer is: deadlock avoidance

Question 54

39. _____ contains all the key parameters about the file system and is read into memory when the computer is booted or the file system is first touched.

Select one:

- a. MBR
- b. Bootblock
- c. Superblock
- d. Partition
- e. None

Feedback

The correct answer is: Superblock

Question 55

4. A search algorithm takes _____ as an input and returns _____ as an output.

Select one:

- a. Input, output
- b. Problem, solution
- c. Solution, problem
- d. None

Feedback

The correct answer is: Problem, solution

Question 56

40. The CPU can request data from an I/O controller one byte at a time, but doing so wastes the CPU's time. What was the solution to this?

Select one:

- a. DMA
- b. Buffering
- c. Partitioning
- d. Parallelizing
- e. All

Feedback

The correct answer is: DMA

Question 57

41. Cascading termination refers to termination of all child processes before the parent terminates, such type of termination is known as terminated _____.

Select one:

- a. Normally
- b. Abnormally
- c. Normally and involuntarily
- d. None of the mentioned

Feedback

The correct answer is: Normally

Question 58

42. Two operations in a schedule are side to be conflict if they satisfy all the following conditions except:

Select one:

- a. They belongs to different transactions
- b. They access the same data item X
- c. At least one of the operation is a write Items(X)
- d. They belongs to the same transactions

Feedback

The correct answer is: They access the same data item X

Question 59

43. Based on a transaction schedule Sb, given as Sb: r1(X); r2(x); w1(X); r1(Y); W2(X); W1(Y); identify non-conflicting operation.

Select one:

- a. r1(X) and w2(X)
- b. r2(X) and w1(X)
- c. w2(X) and w1(Y)
- d. W1(X) and w2(X)

Feedback

The correct answer is: w2(X) and w1(Y)

Question 60

44. Which one of the following locks data items from write but not from read?

Select one:

- a. Implicit lock
- b. Explicit lock
- c. exclusive lock
- d. shared lock

Feedback

The correct answer is: exclusive lock

Question 61

45. Which one of the following Structured Query Languages (SQL) is used to create a database?

Select one:

- a. DML
- b. DDL
- c. DCL
- d. TCL

Feedback

The correct answer is: DDL

Question 62

46. _____ defines a collection (or set) of entities that have the same attributes.

Select one:

- a. entity type
- b. relationship
- c. attribute
- d. constraint

Feedback

The correct answer is: entity type

Question 63

47. The ER model includes all the following modeling concepts except:

Select one:

- a. specialization and generalization
- b. entity
- c. relationship
- d. constraint

Feedback

The correct answer is: specialization and generalization

Question 64

48. Which of the following is a constraint between two sets of attributes from the database?

Select one:

- a. Normalization
- b. functional dependency
- c. Denormalization
- d. All

Feedback

The correct answer is: functional dependency

Question 65

49. Which of the following method is used to enforce access control in a database system based on the granting and revoking privileges?

Select one:

- a. Mandatory Access Control
- b. Role-Based Access Control
- c. Statistical Database Security
- d. Discretionary Access Control
- e. Encryption

Feedback

The correct answer is: Discretionary Access Control

Question 66

5. Breadth first search always expands the _____ node in the problem representation of a search tree.

Select one:

- a. Shallowest
- b. Child node
- c. Deepest
- d. Minimum cost
- e. All
- f. None

Feedback

The correct answer is: Shallowest

Question 67

50. Which of the following is not property of a transaction?

Select one:

- a. Simplicity
- b. B. Atomicity
- c. C. Isolation
- d. D. Durability

Feedback

The correct answer is: Simplicity

Question 68

51. Why the DMBS needs to keep track of operations when transaction starts, terminates and commits or aborts?

Select one:

- a. For concurrency control
- b. For transaction characterization
- c. For recovery purpose
- d. All of the above

Feedback

The correct answer is: For recovery purpose

Question 69

52. The following are benefits of the database systems approach except:

Select one:

- a. Data can be shared
- b. Improved accessibility of data
- c. Security measures can be enforced
- d. Complexity in designing and managing data

Feedback

The correct answer is: Complexity in designing and managing data

Question 70

53. A full scale DBMS should have the following services to be provided to users except:

Select one:

- a. Data storage, retrieval and update in the database
- b. Concurrency Control Services
- c. Recovery Services
- d. Authorization Services (Security)
- e. All of the above

Feedback

The correct answer is: All of the above

Question 71

54. Which one of the following pair of function declaration does not show function overloading?

Select one:

- a. void Abs(); and int Abs();
- b. int Abs(); and int Abs(int);
- c. float Div(float,int); and void Div(long);
- d. float Div(float); and int Div(int);
- e. None of the above

Feedback

The correct answer is: void Abs(); and int Abs();

Question 72

55. Which one of the following array initialization is not possible?

Select one:

- a. int age[3]={1,2,3,4};
- b. float temp[10]={1.2,3.5,1.2};
- c. int age[4]={1,2,3,4};
- d. float temp[]={0.8,0.7,6.4};
- e. None of the above

Feedback

The correct answer is: int age[3]={1,2,3,4};

Question 73

56. If you declare a float pointer as float*ptr, and you declare a float variable as float sum; then which of the following is legal?

Select one:

- a. sum=&ptr;
- b. ptr=∑
- c. *sum=*ptr;

- d. `&sum=ptr;`
- e. All except C

Feedback

The correct answer is: `ptr=∑`

Question 74

57. Which of the following statements are not equivalent, if there is a declaration of an array `a` as follow? `int a[]={1,2,0,3};`

Select one:

- a. `cout<<<a[0];` and `cout<<<a[a[2]];`
- b. `cout<<<a[a[0]+2];` and `cout<<<a[a[0]+a[2]]+a[0];`
- c. `cout<<<a[0];` and `cout<<<a[3]-a[1];`
- d. `cout<<<a;` and `cout<<<a[0];`
- e. None of the above

Feedback

The correct answer is: `cout<<<a;` and `cout<<<a[0];`

Question 75

58. Consider the following code segment `int x=3, y=1; int *p=&x; (*p) ++; p=&y; y-=2;` What is the value of `*p` after the code executes?

Select one:

- a. 3
- b. 1
- c. 2
- d. 4
- e. None of the above

Feedback

The correct answer is: 2

Question 76

59. Assume that you want to declare a function named Model which takes an array of integers and the size of the array as parameters and returns nothing. Which one of the following is the correct syntax of declaring the function?

Select one:

- a. void Model(float [],int);
- b. void Model(int [], int);
- c. void Model(float[],int);
- d. void Model(int a[], int n);
- e. B and D

Feedback

The correct answer is: B and D

Question 77

6. _____is refers to the objective an artificial machine.

Select one:

- a. Initial state
- b. Intermediate state
- c. Goal
- d. Input
- e. None

Feedback

The correct answer is: Goal

Question 78

60. Referring to question number 5, if you want to find the sum of 10 integers stored in an array named list, which one of the following is correct syntax of calling the function?

Select one:

- a. Sum(list[10],10);
- b. Sum(int list[10],10);
- c. void Sum(list,10);
- d. Sum(list,10);
- e. void Sum(int list[10],10);

Feedback

The correct answer is: `Sum(list,10);`

Question 79

61. Which of the following code fragment outputs the number 3 to the screen?

Select one:

- a. `cout << 1 + 2 * 3 / 4 << endl;`
- b. `cout << 1 * 2 + 3 / 4 + 2 << endl;`
- c. `cout << 2 * (2 + 3) / 4 << endl;`
- d. `cout << 1 / 2 * 2 + 3 << endl;`
- e. All of the above

Feedback

The correct answer is: `cout << 1 / 2 * 2 + 3 << endl;`

Question 80

62. Which of the following prototypes does not associate with Prototyping Model?

Select one:

- a. Domain Prototype
- b. Vertical Prototype
- c. Horizontal Prototype
- d. Diagonal Prototype

Feedback

The correct answer is: Diagonal Prototype

Question 81

63. What is the major drawback of the Spiral Model?

Select one:

- a. Higher amount of risk analysis
- b. Doesn't work well for smaller projects
- c. Additional functionalities are added later on
- d. Strong approval and documentation control

Feedback

The correct answer is: Doesn't work well for smaller projects

Question 82

64. When the user participation isn't involved, which of the following models will not result in the desired output?

Select one:

- a. Prototyping & Waterfall
- b. Prototyping & RAD
- c. Prototyping & Spiral
- d. RAD & Spiral

Feedback

The correct answer is: Prototyping & RAD

Question 83

65. Which of the following falls under the category of software products?

Select one:

- a. Firmware, CAD
- b. Embedded, CAM
- c. Customized, Generic
- d. CAD, Embedded

Feedback

The correct answer is: Customized, Generic

Question 84

66. Which of the following activities of the generic process framework delivers a feedback report?

Select one:

- a. Deployment
- b. Planning
- c. Modeling
- d. Construction

Feedback

The correct answer is: Deployment

Question 85

67. Which of the following refers to internal software equality?

Select one:

- a. Scalability
- b. Reusability
- c. Reliability
- d. Usability

Feedback

The correct answer is: Reusability

Question 86

68. Which one of the following activities is not recommended for software processes in software engineering?

Select one:

- a. Software Evolution
- b. Software Verification
- c. Software Testing & Validation
- d. Software designing

Feedback

The correct answer is: Software Verification

Question 87

69. On what basis is plan-driven development different from that of the software development process?

Select one:

- a. Based on the iterations that occurred within the activities.
- b. Based on the output, which is derived after negotiating in the software development process.
- c. Based on the interleaved specification, design, testing, and implementation activities.
- d. All of the above

Feedback

The correct answer is: Based on the iterations that occurred within the activities.

Question 88

7. Which one of the following is a logical connector?

Select one:

- a. ==>
- b. B
- c. C,
- d. D
- e. All
- f. None

Feedback

The correct answer is: All

Question 89

70. How would you store user name 'Informatics' in 'UserName'?

Select one:

- a. setcookie('UserName', 'Informatics');
- b. makeCookie('UserName', 'Informatics');
- c. Cookie('UserName', 'Informatics');
- d. UserName('Informatics');

Feedback

The correct answer is: setcookie('UserName', 'Informatics');

Question 90

71. Select the CSS property that sets the width of an element's bottom border?

Select one:

- a. Border-width
- b. Border-bottom
- c. Border-width-down
- d. Border-buttom-width

Feedback

The correct answer is: Border-bottom-width

Question 91

72. Which of the following is not correct about PHP?

Select one:

- a. Using PHP, you can restrict users to access some pages of your website
- b. PHP can handle forms, i.e. gather data from files, save data to a file, via email you can send data, return data to the user
- c. It depends on the browser and giving the user more control over the browser
- d. Allow you to add, delete, and modify elements within your database via PHP

Feedback

The correct answer is: It depends on the browser and giving the user more control over the browser

Question 92

73. Which one of the following statements should you use to set the session username to INSY3082?

Select one:

- a. `$SESSION['username'] = "INSY3082";`
- b. `$_SESSION['username'] = "INSY3082";`
- c. `session_start("INSY3082");`
- d. `$SESSION_START["username"] = "INSY3082";`

Feedback

The correct answer is: `$_SESSION['username'] = "INSY3082";`

Question 93

74. Which is the correct way to write a JavaScript array?

Select one:

- a. `var txt = new Array(1:"tim",2:"kim",3:"jim")`
- b. `var txt = new Array:1=("tim")2=("kim")3=("jim")`
- c. `var txt = new Array("tim","kim","jim")`
- d. `var txt = new Array="tim","kim","jim"`

Feedback

The correct answer is: `var txt = new Array("tim","kim","jim")`

Question 94

75. When is a cookie saved on a user's web browser?

Select one:

- a. Only when a user submits a form
- b. Whenever a user requests a web page
- c. If a web page is loaded that includes JavaScript code that sets the cookie
- d. If PHP is used to send back an http response header that sets the cookie

Feedback

The correct answer is: If PHP is used to send back an http response header that sets the cookie

Question 95

76. Which one of the following method is responsible for sending the query to the database?

Select one:

- a. `query()`
- b. `send_query()`
- c. `sendquery()`
- d. `query_send()`

Feedback

The correct answer is: `query()`

Question 96

77. Which of the following is the correct way to applying style to a document?

Select one:

- a. Use an external style sheet, either by importing it or by linking to it
- b. Directly embed a document-wide style in the head element of the document
- c. Set an inline style rule using the style attribute directly on an element
- d. All

Feedback

The correct answer is: All

Question 97

78. One of the following is not true about algorithm

Select one:

- a. An algorithm should be independent of programming language
- b. An empirical approach measure reliability of an algorithm
- c. Pseudocode representation of an algorithm is more suitable for analyzing an algorithm
- d. A computer program is an implementation of an algorithm

Feedback

The correct answer is: An empirical approach measure reliability of an algorithm

Question 98

79. An algorithm with a running time $O(\log n)$, where n is an input size, is not faster than an algorithm with running time.

Select one:

- a. $n \log n$
- b. n^2
- c. $2n$
- d. $\log(n/2)$

Feedback

The correct answer is: $\log(n/2)$

Question 99

8. which one of the following is a quantifier?

Select one:

- a. $\Rightarrow$
- b. B
- c. C
- d. D
- e. All
- f. None

Feedback

The correct answer is: C

Question 100

80. An optimization algorithm that tries to find the globally optimal solution to a problem by taking locally optimal solution at each step is ?

Select one:

- a. greedy
- b. dynamic programming
- c. backtracking
- d. divide-and-conquer

Feedback

The correct answer is: greedy

Question 101

81. A running time function of an algorithm whose recurrence relation is represented by $T(n) = 2T(n/2) + n$, where n is an input size and the base condition is $T(1) = 0$

Select one:

- a. $\log n$
- b. n
- c. $n \log n$
- d. $\log n + n$

Feedback

The correct answer is: $n \log n$

Question 102

82. A graph traversal algorithm that is equivalent to the preorder traversal of a tree is?

Select one:

- a. breadth first search
- b. depth first search
- c. best first search
- d. binary search

Feedback

The correct answer is: depth first search

Question 103

83. If a graph has n vertices the minimum spanning tree that can be constructed from the graph will have

Select one:

- a. n edges
- b. $n-1$ vertices
- c. $n-1$ edges
- d. $n+1$ vertices

Feedback

The correct answer is: $n-1$ edges

Question 104

84. the following output of ascendingly sorting a list 5,0,4,1,2 shows that the sorting algorithm most importantly applied is

Select one:

- a. bubble sort
- b. selection sort
- c. insertion sort
- d. quick sort

Feedback

The correct answer is: insertion sort

Question 105

85. One of the following statements is not true about linked lists

Select one:

- a. In singly-linked list navigation can be performed forward only
- b. In doubly-linked list navigation can be done backward only
- c. In singly-circular linked list last node can point to the first node
- d. In any circular linked list there is no any node pointing to NULL

Feedback

The correct answer is: In doubly-linked list navigation can be done backward only

Question 106

86. One of the following is not true about binary tree

Select one:

- a. Binary tree has at most two children
- b. Full binary tree has maximum depth of 2
- c. In balanced binary tree all leaf nodes are at the same level
- d. In a complete binary tree the length from root to any leaf node is h or h-1

Feedback

The correct answer is: Full binary tree has maximum depth of 2

Question 107

87. Which one of the following is not true about a binary search tree constructed from the list of keys 2,1,0,10,7,9,4,3,11,20,15,6,5,8? (if the same order of the list is followed)

Select one:

- a. The tree has depth 5
- b. The keys 4 and 9 are at the same level
- c. The deepest node has the key 5
- d. The tree is a full binary tree

Feedback

The correct answer is: The tree is a full binary tree

Question 108

88. Which feature of OOP is implemented by the following code? `class Student {int marks;} class Postgraduate extends Student { int age; public Postgraduate (int age){ this.age=age; } }`

Select one:

- a. Encapsulation and Inheritance
- b. Inheritance and polymorphism
- c. Polymorphism

d. Inheritance

Feedback

The correct answer is: Inheritance

Question 109

89. All are true about polymorphism, except?

Select one:

- a. It is an object-oriented concept by which we can perform a single action in different ways.
- b. Polymorphism occurs when a parent class reference is used to refer a child class object.
- c. Any Java object that can pass more than one IS-A test is considered to be polymorphic.
- d. Polymorphism occurs when a child class reference is used to refer a parent class object.
- e. None of the above

Feedback

The correct answer is: Polymorphism occurs when a child class reference is used to refer a parent class object.

Question 110

9. Which string is not belongs to the regular expression $(0+1)^*1$?

Select one:

- a. 1111
- b. 0000
- c. 1110
- d. 1010
- e. All except A.

Feedback

The correct answer is: All except A.

Question 111

90. Suppose that you have given a super class named Animal. The class Animal contains only parametrized constructors and you also have defined Animal's subclass called Dog. The Dog class is the demonstration class that contains the main method. Based on the given scenario, which of the following is/are false statement /s?

Select one:

- a. it is possible to create an object of Animal within the main method of the subclass with the syntax `Animal animal1 = new Animal ();`
- b. It is a must to define a constructor with in the subclass Dog
- c. It is a must to use a super () call in the subclass constructors to invoke the super class constructors.
- d. There is no default constructor provided by the compiler for the Animal class.
- e. All except A

Feedback

The correct answer is: it is possible to create an object of Animal within the main method of the subclass with the syntax `Animal animal1 = new Animal ();`

Question 112

91. We use this keyword in java. All are true about this keyword except?

Select one:

- a. We use this keyword to refer the current object instance data members of its own class.
- b. We use this keyword to invoke the current objects' instance method of its own class.
- c. We use this keyword to invoke the constructor of own class from within another constructor of the same class
- d. We use this keyword to invoke the superclass constructor from the subclass constructor

Feedback

The correct answer is: We use this keyword to invoke the superclass constructor from the subclass constructor

Question 113

92. What is encapsulation in object-oriented programming?

Select one:

- a. It is a way of combining various data members and member functions that operate on those data members into a single unit
- b. It is a way of combining various data members and member functions into a single unit which can operate on any data
- c. It is a way of combining various data members into a single unit
- d. It is a way of combining various member functions into a single unit

Feedback

The correct answer is: It is a way of combining various data members and member functions that operate on those data members into a single unit

Question 114

93. Assume that you are attempting to create a non-inheritable Lion class in java. Therefore, which of the following will define the class correctly?

Select one:

- a. `class Lion { //.... }`
- b. `final abstract class Lion { //..... }`
- c. `static final abstract class Lion { //..... }`
- d. `final class Lion { //..... }`

Feedback

The correct answer is: `final class Lion { //..... }`

Question 115

94. suppose that you are given the following java code segment
`abstract class Animal { public abstract void sound (); public void eat() { System.out.println("All animals can eat!"); } } public class Dog extends Animal { }.` Therefore, what is possible in the Dog subclass?

Select one:

- a. Creating an object of Animal in the main method as `Animal a = new Animal ()`
- b. Overriding the `sound ()` method in the subclass is must unless the Dog class is defined itself as an abstract
- c. Overriding the `eat ()` method in the subclass as `void eat () { //.... }`
- d. It is possible to create an object as `Animal animal1 = new Dog ()`
- e. B and D
- f. A and C

Feedback

The correct answer is: B and D

Question 116

95. Which of the following is correct way of interface definition?

Select one:

- a. `public interface Book { int bookNo; String title; void setBookNo( int bookNo) { This.bookNo = bookNo;} }`
- b. `public interface Book { int bookNo; String title; void setBookNo (int bookNo); }`
- c. `final public interface Book { int bookNo; String title; void setBookNo (int bookNo) { //.....}; }`
- d. None of the above

Feedback

The correct answer is: `public interface Book { int bookNo; String title; void setBookNo (int bookNo); }`

Question 117

96. What is true about abstract class?

Select one:

- a. An abstract class are not complete by their own
- b. An abstract class cannot instantiate an object of its own
- c. An abstract class must have at least one abstract method
- d. An abstract class can implement another abstract class methods
- e. All of the above

Feedback

The correct answer is: All of the above

Question 118

97. Assume that you are given two interfaces named A, and B and one class named C to implement the two interfaces, therefore, which of the following is correct class definition?

Select one:

- a. class C implements A, B {}
- b. class C implements A, implements B {}
- c. class C extends A, B {}
- d. class C extends A, extends B {}

Feedback

The correct answer is: class C implements A, B {}

Question 119

98. Assume that you have defined only parametrized constructor in your employee class as given below as a segment of program. public class Employee { String name; float salary Employee (String name, float salary) { this. salary = salary; this.name = name; } } So, which of the following statements are invoking the constructor properly?

Select one:

- a. Employee emp = new Employee ();
- b. new Employee ();
- c. Employee emp = new Employee ("Dawit", 13,500.00f);
- d. new Employee ("Dawit", 13,500.00f);
- e. C and D

Feedback

The correct answer is: C and D

Question 120

99. Which one of the following is not among the major activities of RSA cryptosystem?

Select one:

- a. Signature verification
- b. Key generation
- c. Digital signing
- d. Message digest

Feedback

The correct answer is: Digital signing